Eshaan Marocha

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Education

University of Toronto Sep. 2024 – Apr. 2029
Bachelor of Applied Science – Electrical Engineering + PEY Co-op *Toronto, ON*

- GPA: 3.93/4.0 (90%+ average)
- Minor: Artificial Intelligence
- Relevant Courses: Digital Systems (Quartus Prime, ModelSim, Verilog), Circuit Analysis (LTspice), Hardware Design and Communication (Altium Designer), Programming Fundamentals (C/C++)

Skills

HDLs & Programming: Verilog, SystemVerilog, Python, C, C++
Design & Simulation Tools: LTspice, Vivado, Quartus Prime, ModelSim, Altium Designer, KiCad, Fusion 360
Digital Design Concepts: RTL Design, FPGA Development, ASIC Design, Finite State Machines, Embedded Systems
AI & Software: Computer Vision, OpenCV, Google Gemini API, Firebase/Firestore
Soft Skills: Collaboration, Communication, Problem Solving, Critical Thinking, Attention to Detail

Work Experience

University of Toronto Intelligent Sensory Microsystems Laboratory May 2026 – Present
FPGA Design Intern *Toronto, ON*

- Modified camera controller RTL using Vivado so external signals could trigger exposures in sync with a laser projector, enabling depth reconstruction within images.
- Extended camera controller RTL with a double-buffered memory, so per-pixel exposure settings could be uploaded from Python and swapped mid-capture without pausing the camera.
- Designed a SystemVerilog AXI-Stream bridge for bidirectional data transfer between a Python host and an FPGA, sustaining over 200 MB/s with no dropped data.
- Designed a 4-layer interposer PCB in Altium Designer to break out and level-shift 75 MHz signals from an existing board, making them accessible to an FPGA.
- Optimized the camera's image capture pipeline by eliminating two noise sources traced to exposure settings assumed to be irrelevant, increasing signal-to-noise ratio by approximately 2.9×.

Excel Drafting Inc. May 2025 – Aug. 2025
Software Automation Intern *Acton, ON*

- Created company-wide macros for automatic project property updates, reducing manual editing time by 93%.
- Automated bill of material formatting using a Python script, eliminating formatting errors.

Projects

Ray Tracing ASIC Feb. 2026 – Aug. 2026
IC Design Hackathon, IEEE *Toronto, ON*

- Designed a ray-tracing ASIC in SystemVerilog that accelerates the rendering of 3D scenes into 2D images with realistic lighting, shadows, and reflections.
- Architected a software-hardware-software pipeline that preprocesses 3D models in Python, streams data to the chip over QSPI for the heavy computation, and returns results to Python for final rendering.
- Took the design through automated synthesis, place-and-route, and signoff using the TinyTapeout SKY130 flow, closing timing at 40 MHz with zero setup or hold violations and passing DRC, LVS, and antenna checks.
- Fit the full design into a 4×2 tile area (149,000 μm^2, 24,800 cells) drawing 5.8 mW by making power, performance, and area aware decisions such as a step limit bounding the worst-case work per ray.
- Verified correctness at both the module and full-chip level using reference models that compute expected results, combining directed edge-case tests with large randomized tests (10,000+ inputs).
- Implemented the design on an Artix-7 FPGA at 100 MHz, rendering full 4K images in under 60 seconds.

Muscle Controlled Robotic Arm Mar. 2026 – Apr. 2026
Computer Organization Course Project, U of T *Toronto, ON*

- Built a 5-degree-of-freedom robotic arm that translates human muscle flexion into real-time motion, using an electromyography sensor to detect electrical activity from muscle activation.
- Designed and 3D-modelled the robotic arm in Fusion 360, then 3D printed, assembled, and wired the full system including five servo motors running on a 6 V power rail.
- Developed embedded C software on a Nios V processor running on an FPGA to detect muscle flexion, switch between control modes, and generate 50 Hz pulse-width modulation signals to drive the five servos in real time.
- Debugged hardware integration issues including an unusable analog-to-digital converter port, wire-management failures, and a burnt out gripper servo, enabling a working final demo within a 3-week build timeline.

HF Radio Transceiver Jan. 2026 – Apr. 2026
Hardware Design and Communication Course Project, U of T *Toronto, ON*

- Designed a power amplifier and filter in Altium for an HF radio transceiver covering the 8-16 MHz band.
- Implemented a class D amplifier achieving 61% efficiency while delivering 2 W into a standard 50 Ω antenna load.
- Designed a 5-pole output filter to clean up the amplifier's switching waveform, achieving 0.73% total harmonic distortion (THD) at 14 MHz as measured using Python-automated instrument sweeps.

Cupid Glasses Feb. 2026
Make U of T 2026 Hackathon, IEEE *Toronto, ON*

- Won 1st Place and Best Use of Google Gemini API at MakeUofT 2026 for wearable smart glasses that identify who the wearer is speaking with and display personalized icebreakers on an LCD, built in 24 hours.
- Streamed live video from an ESP32-CAM mounted on the glasses to a Python backend over a local WiFi network to enable server-side data processing.
- Identified the person in frame by processing incoming frames with OpenCV and matching them against stored profile photos using the DeepFace recognition model.
- Queried Firebase to pull the matched person's interests, then sent interest context to the Google Gemini API to generate tailored icebreakers to be displayed on the LCD.

FPGA Drum Pad Nov. 2025
Digital Systems Course Project, U of T *Toronto, ON*

- Built a 12 pad electronic drum controller on a DE1-SoC FPGA in Verilog, with recording, playback, and metronome modes.
- Designed a sound engine that streams 16-bit samples from 12 on-chip ROMs consuming roughly 70% of the FPGA's block RAM, mixing all 12 sounds each sample period through an adder tree sized to guarantee no overflow.
- Conditioned 19 asynchronous inputs from 12 drum pads, a rotary encoder, and action buttons using multi-stage synchronizers and per-input debounce counters, preventing metastability and false triggers.
- Verified and refined the design by simulating modules in ModelSim, synthesizing in Quartus, and iteratively testing on hardware, ensuring the final system behaved as intended.

Stock Trading Algorithm May 2025 – Jun. 2025
Personal Project *Toronto, ON*

- Developed a Python (+ NumPy and Pandas) based automated stock-trading strategy that scans all S&P 500 stocks and generates buy/sell signals.
- Evaluated the strategy by building a backtesting engine and testing it on real S&P 500 price data from Jun. 2024 to Jun. 2025, achieving a 24.8% compound annual growth rate (CAGR) over that period.
- Implemented visual performance analytics by plotting equity curves and benchmark comparisons to the S&P 500 with Matplotlib, making it easy to evaluate the strategy over time.

Awards

NSERC Undergraduate Student Research Award | *University of Toronto* Mar. 2026
1st Place in Valentines Theme Make U of T 2026 | *IEEE U of T* Feb. 2026
Best use of Gemini API Make U of T 2026 | *IEEE U of T* Feb. 2026
3rd Place Overall Hack Without Borders 2025 | *Engineers Without Borders* Mar. 2025
Walter Scott Guest Memorial Admission Scholarship | *University of Toronto* Sep. 2024
Edward S. Rogers Sr. Admission Scholarship | *University of Toronto* Sep. 2024